CSE 457
Machine Learning

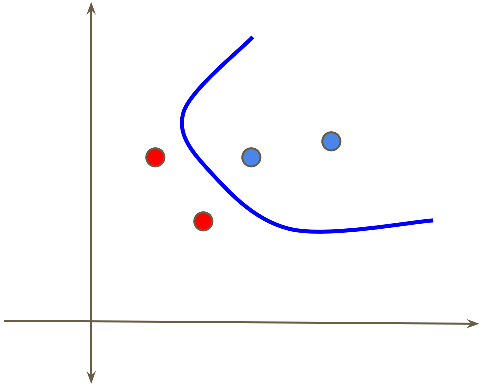
Support Vector Machines
Fall 2025

Contents

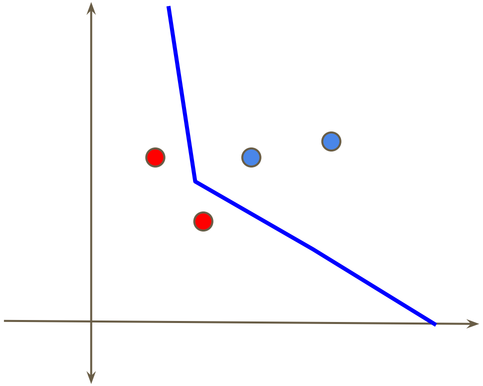
- The idea of SVM
- Kernel Functions
- Bias vs Variance
- Testing a ML Model

https://www.youtube.com/watch?v=_PwhiWxHK8o

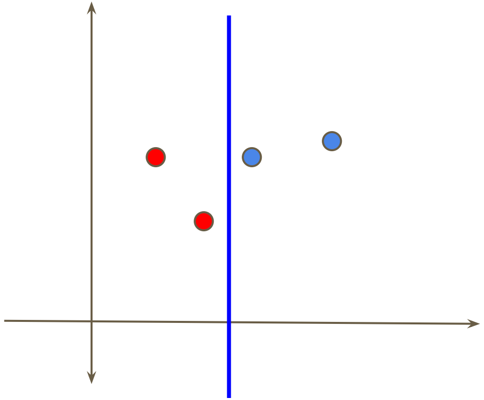
Decision Boundaries



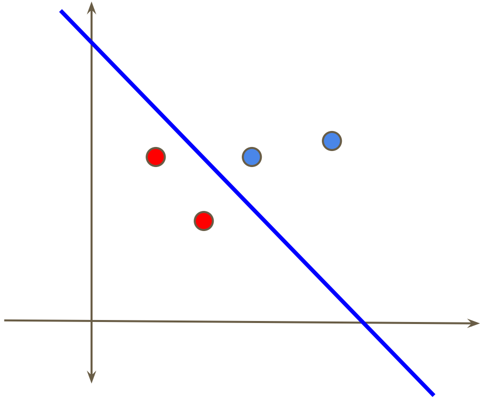
Neural Network



KNN



Decision Tree



Logistic Regression

Idea from - Vladimir Vapnik

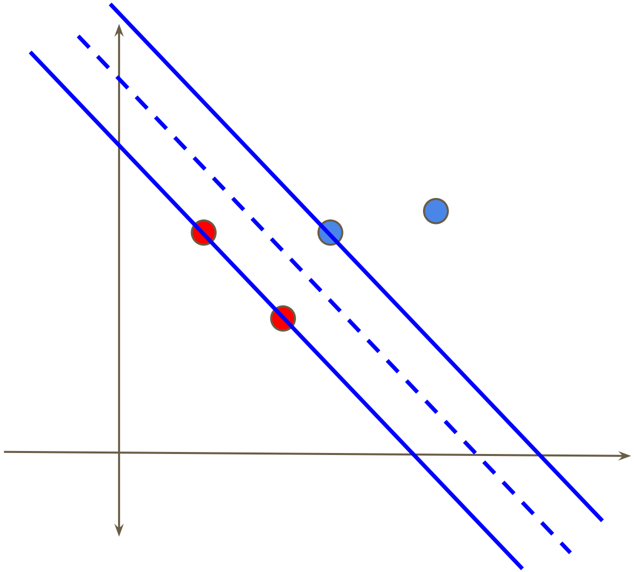
Think of a street that separates them

The widest street possible!

Widest street approach!

What else you need?

- We need a decision rule
- We need to learn the parameters of that street

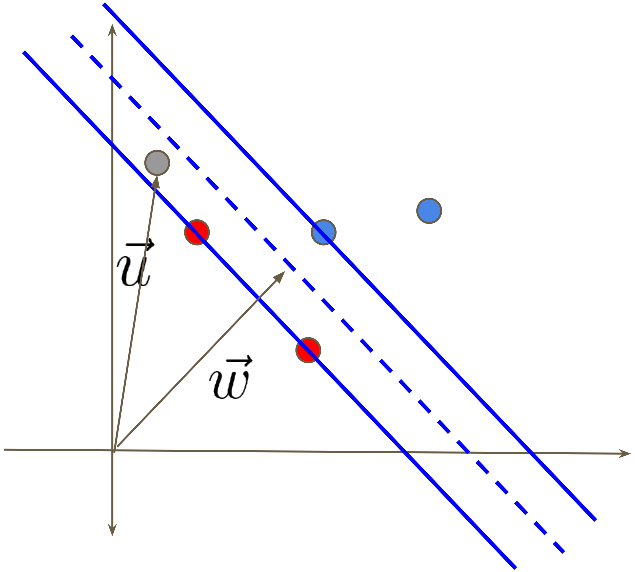


Decision Rule

Let **w** be the vector directing to the middle of the street and perpendicular!

Let **u** be the vector denoting an unknown instance!

We have to decide whether **u** is on the right side or the left side of the street



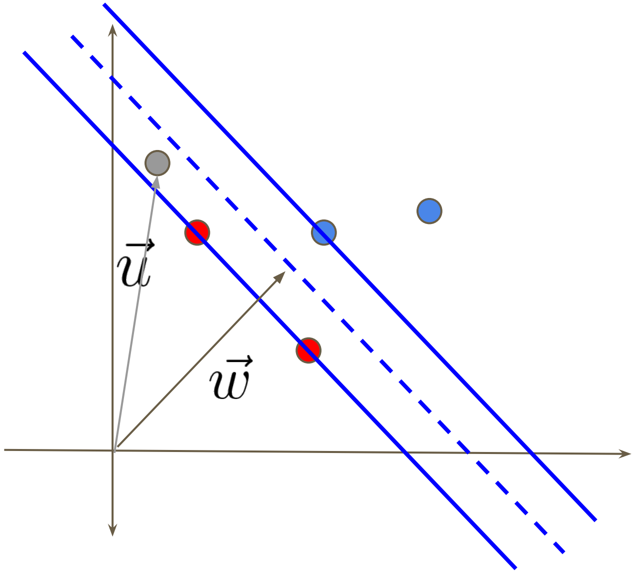
Decision Rule

We have got **w** and **u**, two vectors.

Now we can get the projection of u in the direction of **w**

The projection will give us an idea how far it is from the middle of the road.

Projection is simple: dot product of the vectors **w . u**



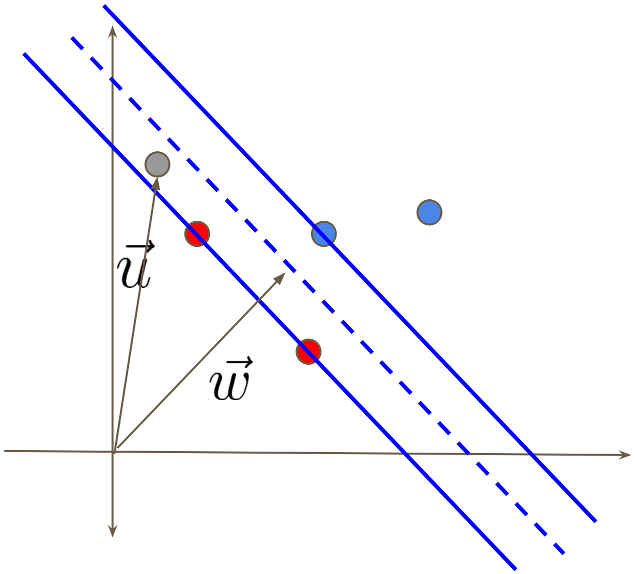
Decision Rule

We check whether this projection is greater than a constant c.

w.v >= c $\vec{w}.\vec{u} \geq c$

If it crosses the median line of the street, positive class or blue class, else red class or negative class.

c = -b



If **w.u** +b >= 0
    label = +1
else
    Label = -1

Decision Rule - SVM

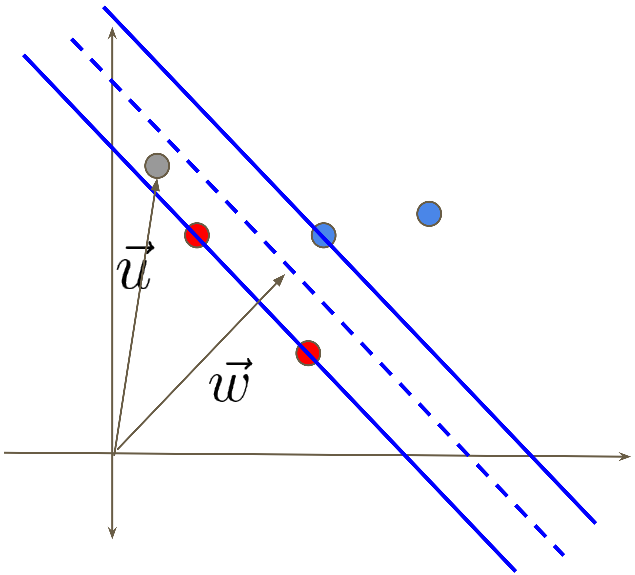
If **w.u** +b >= 0
 label = +1
else
 Label = -1

We do not know **w**

We do not know b

These are the parameters that we need to learn for SVM

We just know that w is perpendicular to the median line



Support Vector Machines

If **w.u** +b >= 0

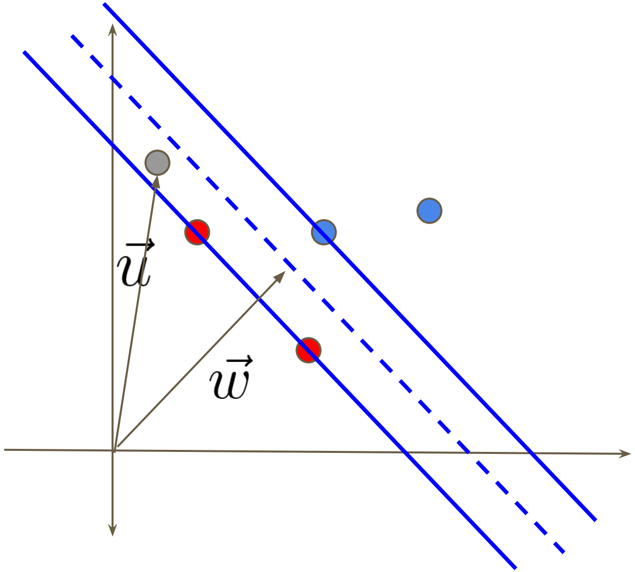
label = +1

else

Label = -1

We assume that positive and negative samples are constrained. i.e. the dot products will give a value greater or equal to +1 for positive and less or equal to -1 for negative samples.

$$\vec{w}.\vec{x}_+ + b \geq 1$$
$$\vec{w}.\vec{x}_- + b \leq -1$$

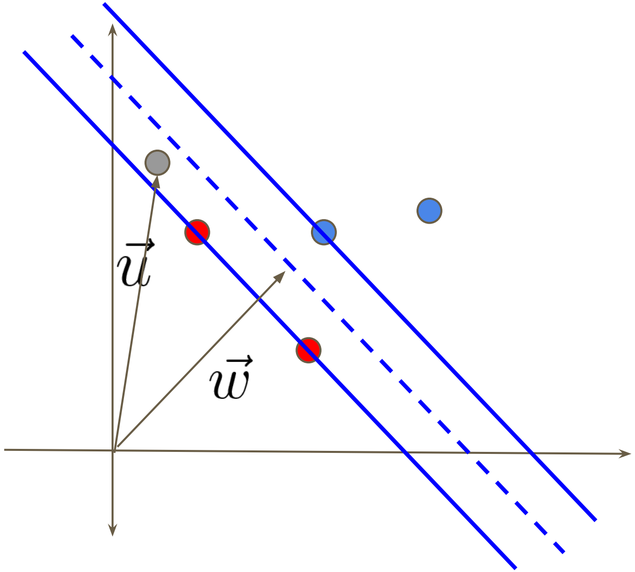


Support Vector Machines

Now, here we have two inequalities. One for the positive samples and another for the negative samples. We wish to generalize them. We need another variable to help. Lets call that **y**. Yes its the labels. y = +1 for the positive samples, -1 for the negative samples

$$\vec{w}.\vec{x}_+ + b \geq 1$$
$$\vec{w}.\vec{x}_- + b \leq -1$$

If **w.u** +b >= 0
    label = +1
else
    Label = -1



Support Vector Machines

If **w.u** +b >= 0

label = +1

else

Label = -1

Lets multiply both the equations with y

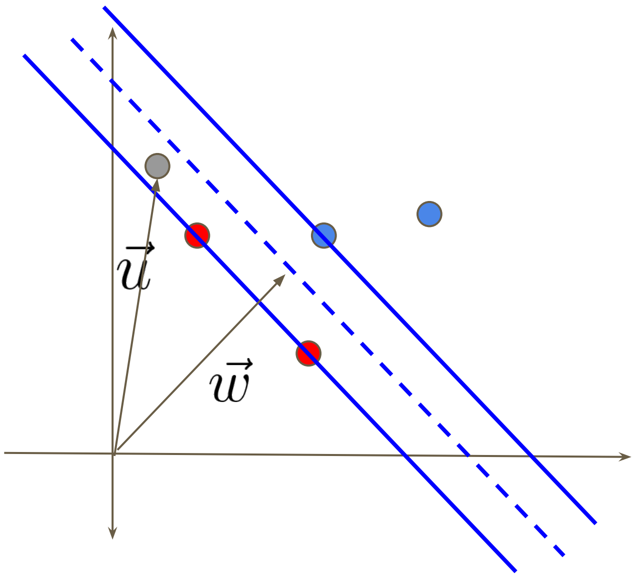
$$\vec{w}.\vec{x}_+ + b \geq 1$$
$$\vec{w}.\vec{x}_- + b \leq -1$$

$$\Rightarrow$$

$$y_i(\vec{w}.\vec{x}_i + b) \geq 1$$
$$y_i(\vec{w}.\vec{x}_i + b) \geq 1$$

$$\Downarrow$$

$$y_i(\vec{w}.\vec{x}_i + b) - 1 \geq 0$$

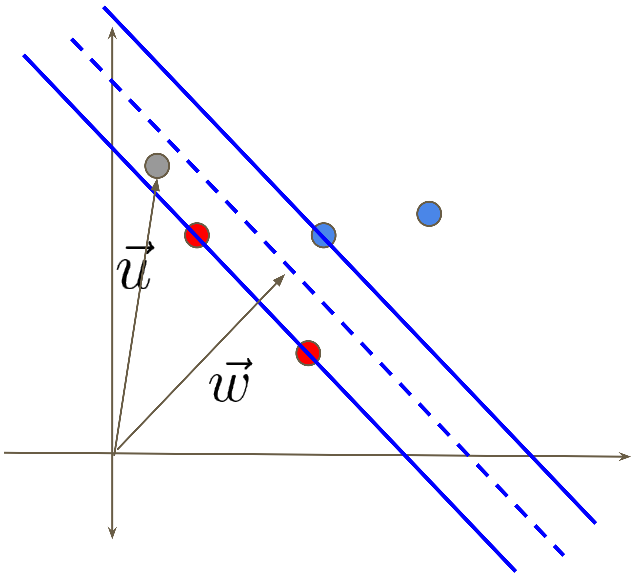


Support Vector Machines

For the support vectors or the **x** that are on the gutter,

$$y_i(\vec{w}.\vec{x}_i + b) - 1 = 0$$

If **w.u** +b >= 0
    label = +1
else
    Label = -1



Is this the magic moment?

Wait, remember the widest street is not yet there!

Width of the steet = distance between the support vectors opposite to each other on the street?

Support Vector Machines

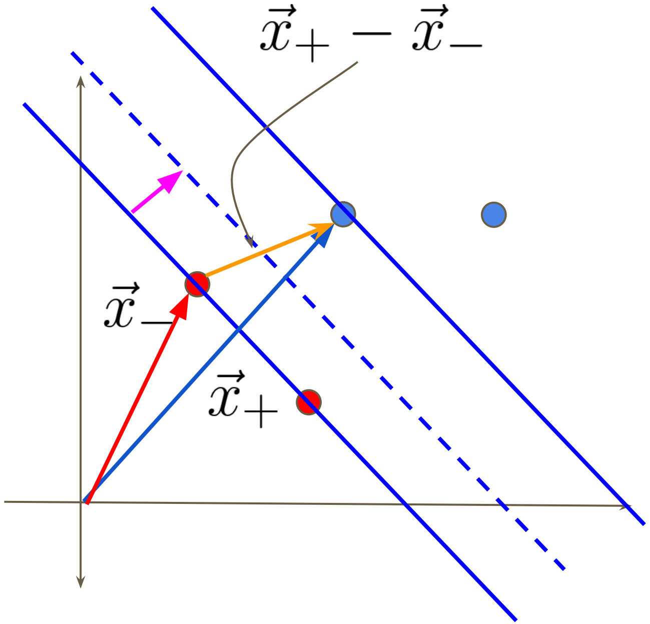
Lets have positive and negative support vectors: $\mathbf{x}_+$ $\mathbf{x}_-$

Also lets have a unit vector to the direction perpendicular to the median line

w is a vector in that direction, we can get a unit normal vector by dividing it by its magnitude

Then the project of the minus vector in that direction is the width of the street!

If **w.u** +b >= 0
    label = +1
else
    Label = -1



Support Vector Machines

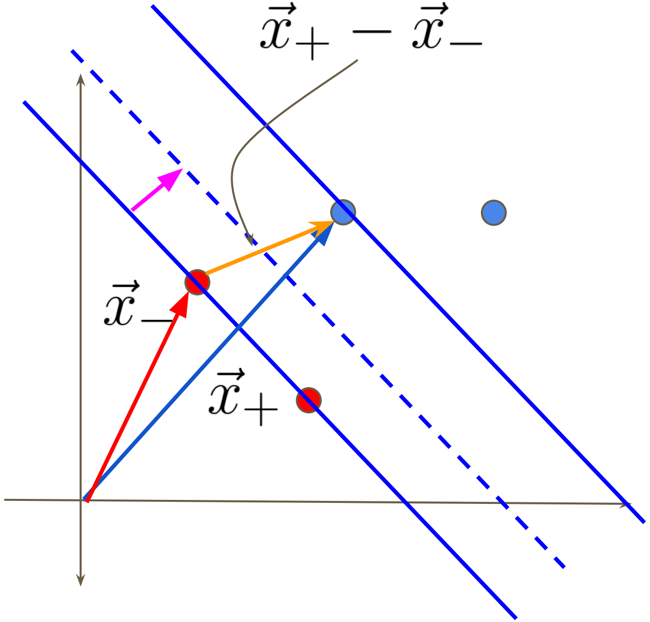
If **w.u** +b >= 0
    label = +1
else
    Label = -1

Lets have positive and negative support vectors: $\mathbf{x}_+$ $\mathbf{x}_-$

w is a vector in that direction, we can get a unit normal vector by dividing it by its magnitude

Then the project of the minus vector in that direction is the width of the street!

$$width = (\vec{x}_+ - \vec{x}_-) . \frac{\vec{w}}{||\vec{w}||}$$



Support Vector Machines

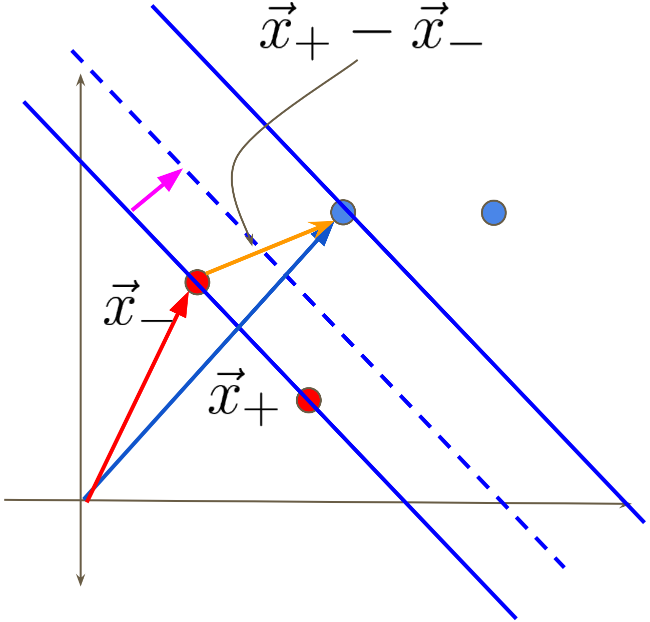
If **w.u** +b >= 0
label = +1
else
Label = -1

$$y_i(\vec{w}.\vec{x}_i + b) = 1$$

$$width = (\vec{x}_+ - \vec{x}_-).\frac{\vec{w}}{||\vec{w}||}$$

$$width = (\underbrace{\vec{x}_+.\vec{w}}_{1-b} - \underbrace{\vec{x}_-.\vec{w}}_{1+b}).\frac{1}{||\vec{w}||}$$

$$width = \frac{2}{||\vec{w}||}$$



Support Vector Machines

If **w.u** +b >= 0
label = +1
else
Label = -1

We want to maximize this width

$$\text{maximize } \frac{2}{||\vec{w}||}$$

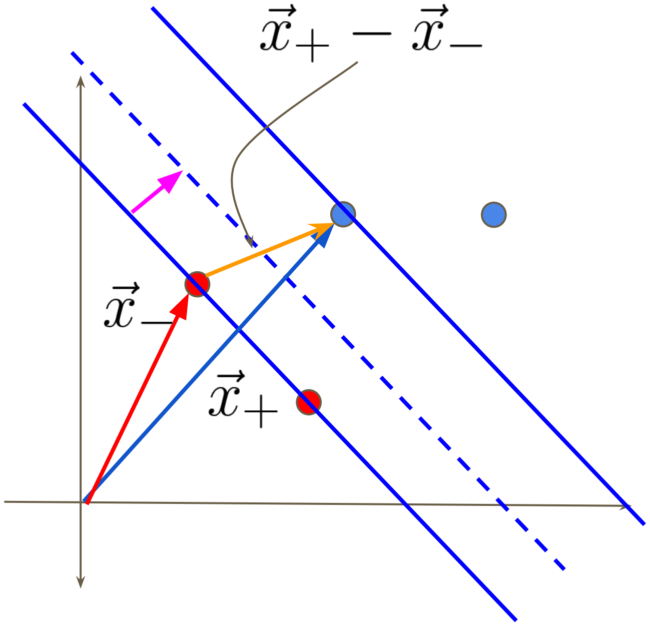
$$\Downarrow$$

$$\text{maximize } \frac{1}{||\vec{w}||}$$

$$\Downarrow$$

$$\text{minimize } ||\vec{w}||$$

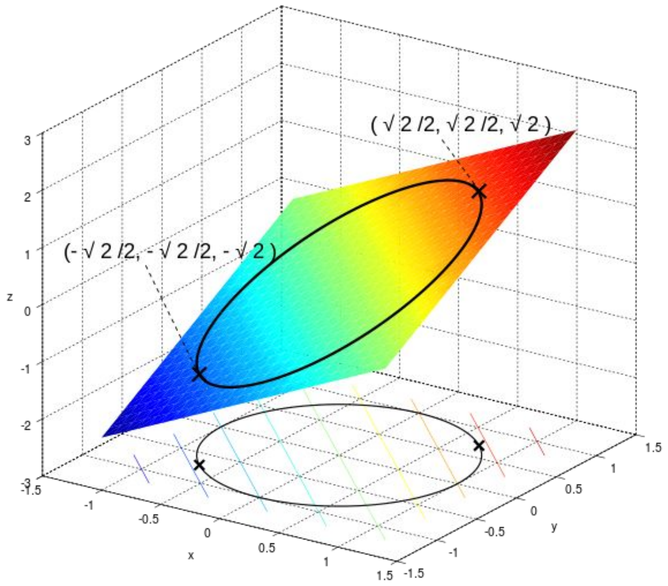
$$\boxed{\text{minimize } \frac{1}{2}||\vec{w}||^2}$$



Constrained Optimization

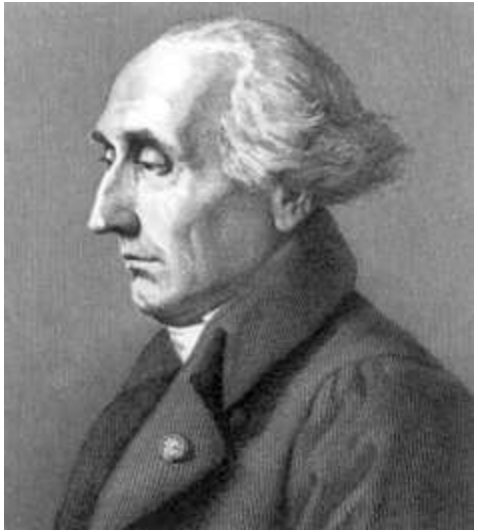
Optimize f(x) with constraints c(x), d(x), ….

Suppose we wish to maximize $f(x,y)=x+y$ subject to the constraint $x^{2}+y^{2}=1$

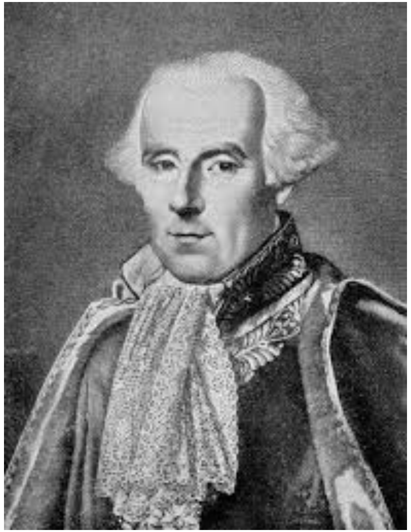


https://en.wikipedia.org/wiki/Lagrange_multiplier

Fourier - A sad student or a genius?



Lagrange



Laplace

Legendre

Fourier

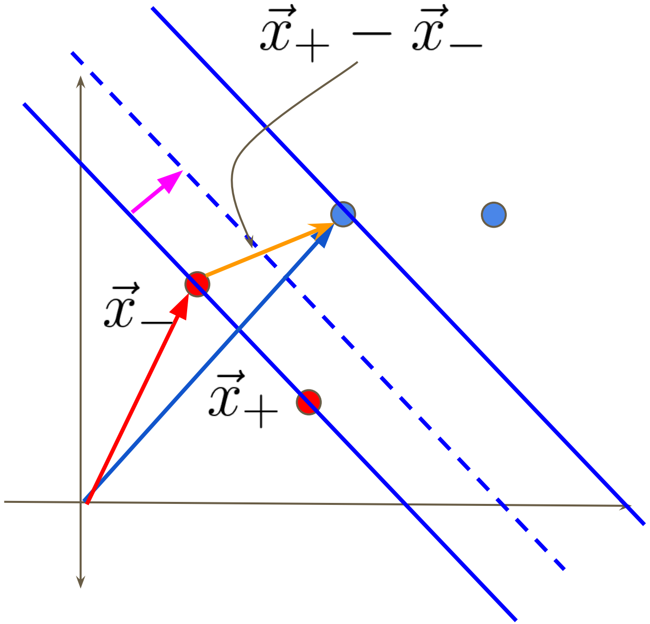
Lagrange Multiplers

If **w.u** +b >= 0
label = +1
else
Label = -1

$$\boxed{\text{minimize } \frac{1}{2}||\vec{w}||^2}$$

$$y_i(\vec{w}.\vec{x}_i + b) - 1 = 0$$

$$\mathcal{L} = \frac{1}{2}||\vec{w}||^2 - \sum \alpha_i[y_i(\vec{w}.\vec{x}_i + b) - 1]$$



Lagrange Multiplers

$$\boxed{\text{minimize } \frac{1}{2}||\vec{w}||^2}$$

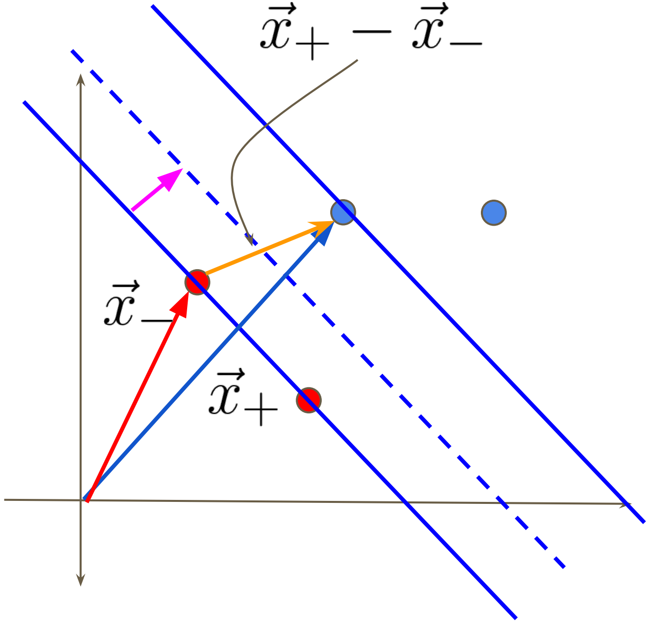
$$y_i(\vec{w}.\vec{x}_i + b) - 1 = 0$$

If **w.u** +b >= 0
label = +1
else
Label = -1

$$\mathcal{L} = \frac{1}{2}||\vec{w}||^2 - \sum \alpha_i[y_i(\vec{w}.\vec{x}_i + b) - 1]$$

$$\frac{d\mathcal{L}}{d\vec{w}} = \vec{w} - \sum \alpha_i y_i \vec{x}_i = 0$$

$$\vec{w} = \sum \alpha_i y_i \vec{x}_i$$



Lagrange Multiplers

$$\text{minimize } \frac{1}{2}||\vec{w}||^2$$

$$y_i(\vec{w}.\vec{x}_i + b) - 1 = 0$$

If **w.u** +b >= 0
label = +1
else
Label = -1

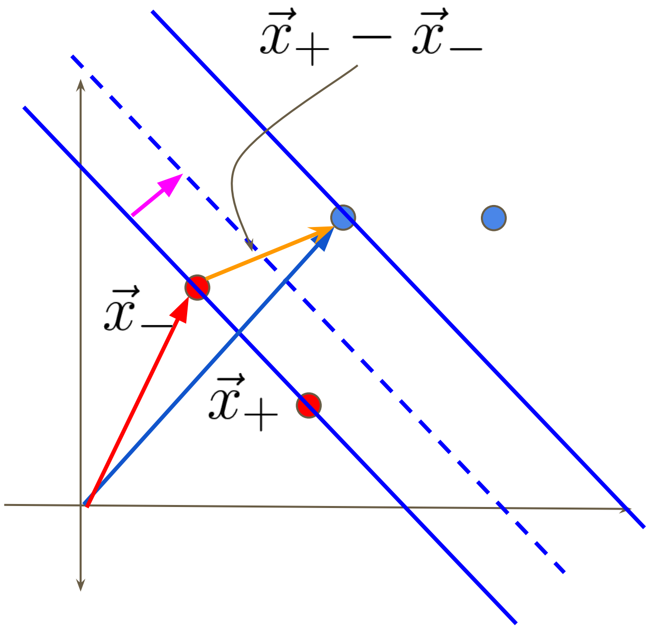
$$\boxed{\mathcal{L} = \frac{1}{2}||\vec{w}||^2 - \sum \alpha_i[y_i(\vec{w}.\vec{x}_i + b) - 1]}$$

$$\frac{d\mathcal{L}}{d\vec{w}} = \vec{w} - \sum \alpha_i y_i \vec{x}_i = 0 \Rightarrow \vec{w} = \sum \alpha_i y_i \vec{x}_i$$

$$\frac{d\mathcal{L}}{db} = -\sum \alpha_i y_i = 0 \Rightarrow \sum \alpha_i y_i = 0$$

$$\mathcal{L} = \frac{1}{2}(\sum \alpha_i y_i \vec{x}_i)(\sum \alpha_i y_i \vec{x}_i) - \sum \alpha_i y_i \vec{x}_i.(\sum \alpha_i y_i \vec{x}_i) - \sum \alpha_i y_i b + \sum \alpha_i$$

$$\mathcal{L} = \sum \alpha_i - \frac{1}{2} \sum \sum \alpha_i \alpha_j y_i y_j \boxed{\vec{x}_i.\vec{x}_j}$$



Support Vector Machines

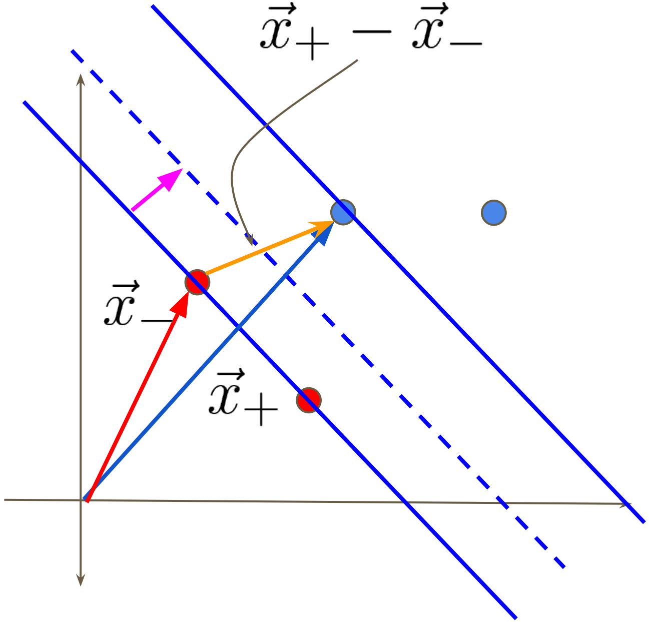
$$\text{minimize } \frac{1}{2}||\vec{w}||^2$$

$$\vec{w} = \sum \alpha_i y_i \vec{x}_i$$

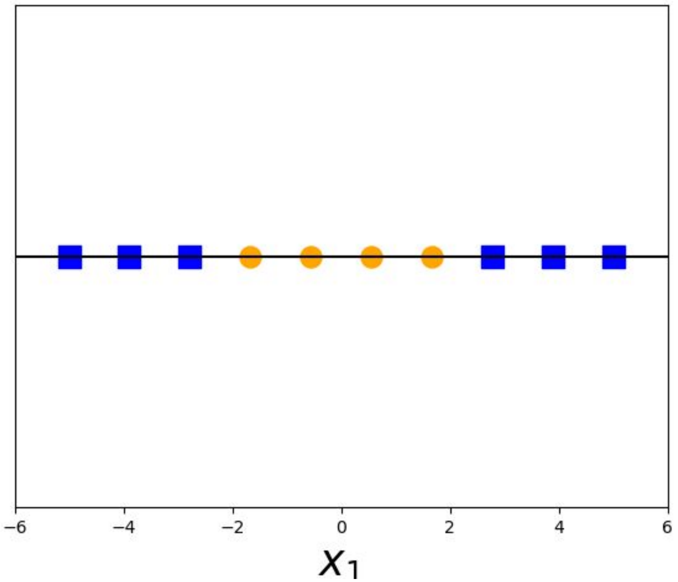
If **w.u** +b >= 0
label = +1
else
Label = -1

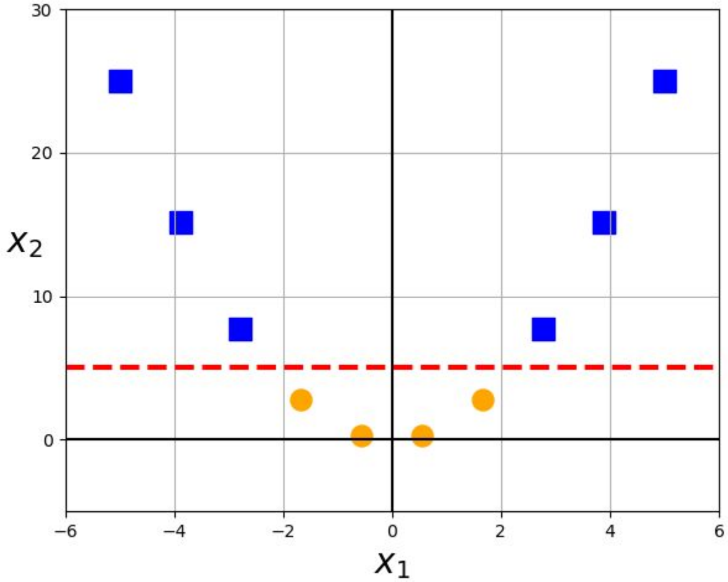
$$\text{if } \sum \alpha_i y_i \boxed{\vec{x}_i . \vec{u}} + b \geq 0 \text{ label} = +$$
$$\text{else label} = -$$

It is a dot product

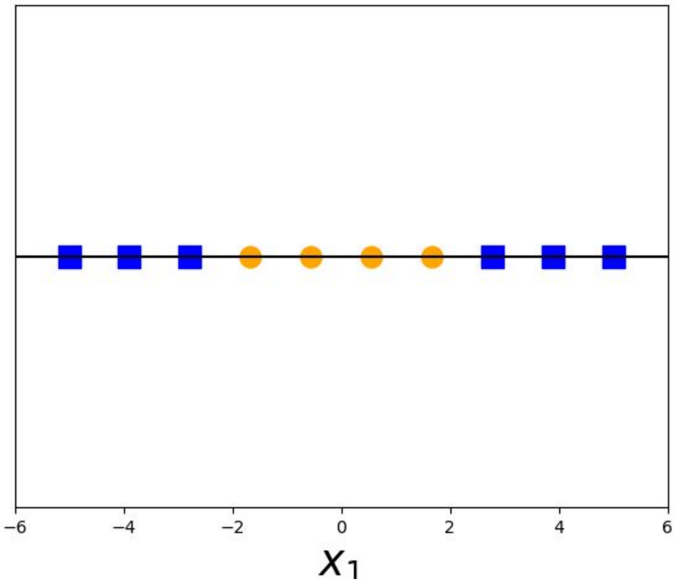


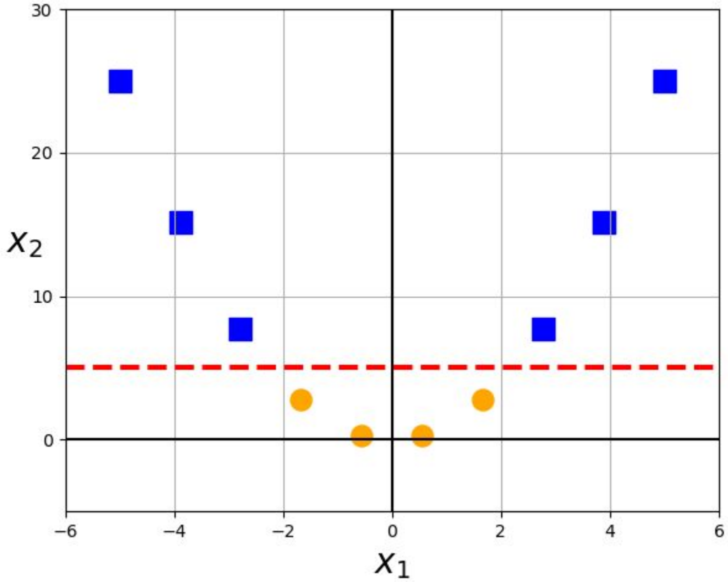
Support Vector Machines - The kernel trick



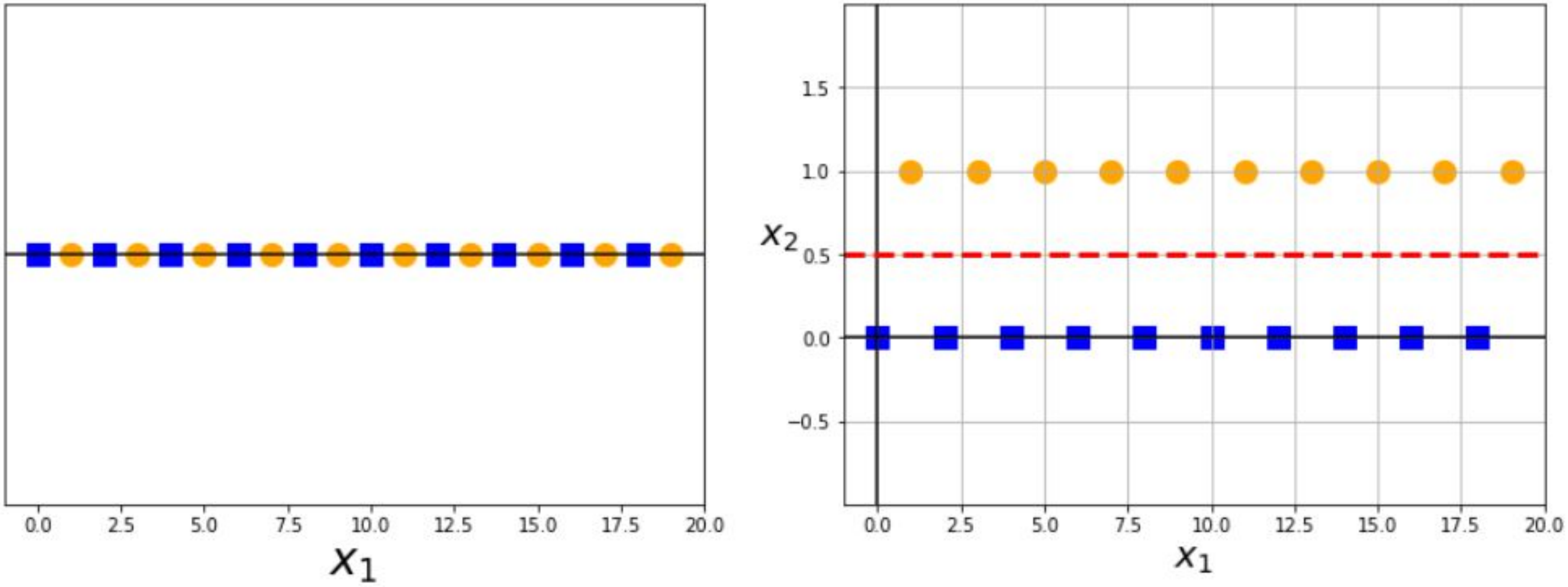


The kernel trick (quadratic)





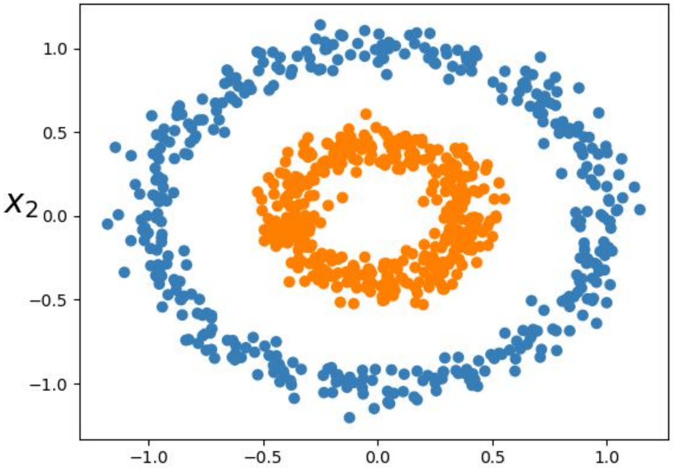
The kernel trick (mod 2)

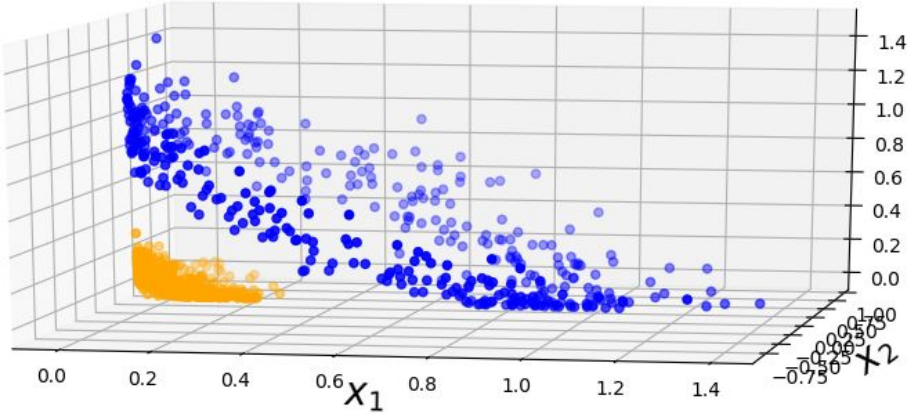


The kernel trick (polinomial)

Equation 5-8. Second-degree polynomial mapping

$$\phi(\mathbf{x}) = \phi\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} {x_1}^2 \\ \sqrt{2}\, x_1 x_2 \\ {x_2}^2 \end{pmatrix}$$





The kernel trick

We do not need a mapping $\phi()$ to convert all the input vector

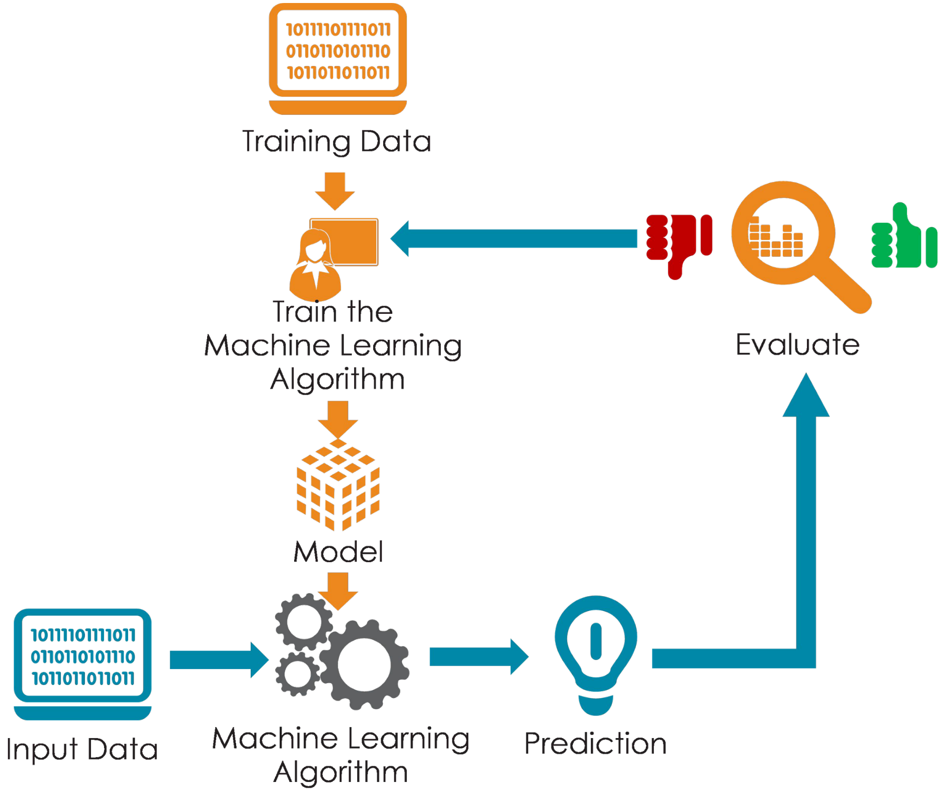
We need a kernel $K(x_i,x_j)=\phi(x_i) \cdot \phi(x_j)$

There are many

- Linear
- Polynomial
- Radial basis
- Sigmoid

How to test you model?

- Evaluation Metrics?
  - Accuracy vs Error
  - Thresholds
- Sampling
  - Separate test set
  - Cross-Fold Validation
  - Leave one out



Regression

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

MSE = mean squared error

n = number of data points

Y_i = observed values

$\hat{Y}_i$ = predicted values

$$\text{MAE} = \frac{\sum_{i=1}^{n}|y_i - x_i|}{n}$$

MAE = mean absolute error

y_i = prediction

x_i = true value

n = total number of data points

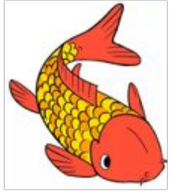
Heavily punishes outliers

Binary Classification - Confusion Matrix

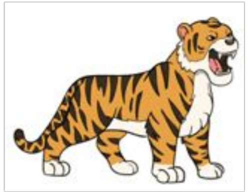
- **True positive:** Sick people correctly identified as sick
- **False positive:** Healthy people incorrectly identified as sick
- **True negative:** Healthy people correctly identified as healthy
- **False negative:** Sick people incorrectly identified as healthy

		ACTUAL	
		Negative	Positive
PREDICTION	Negative	TRUE NEGATIVE	FALSE NEGATIVE
	Positive	FALSE POSITIVE	TRUE POSITIVE

Examples?

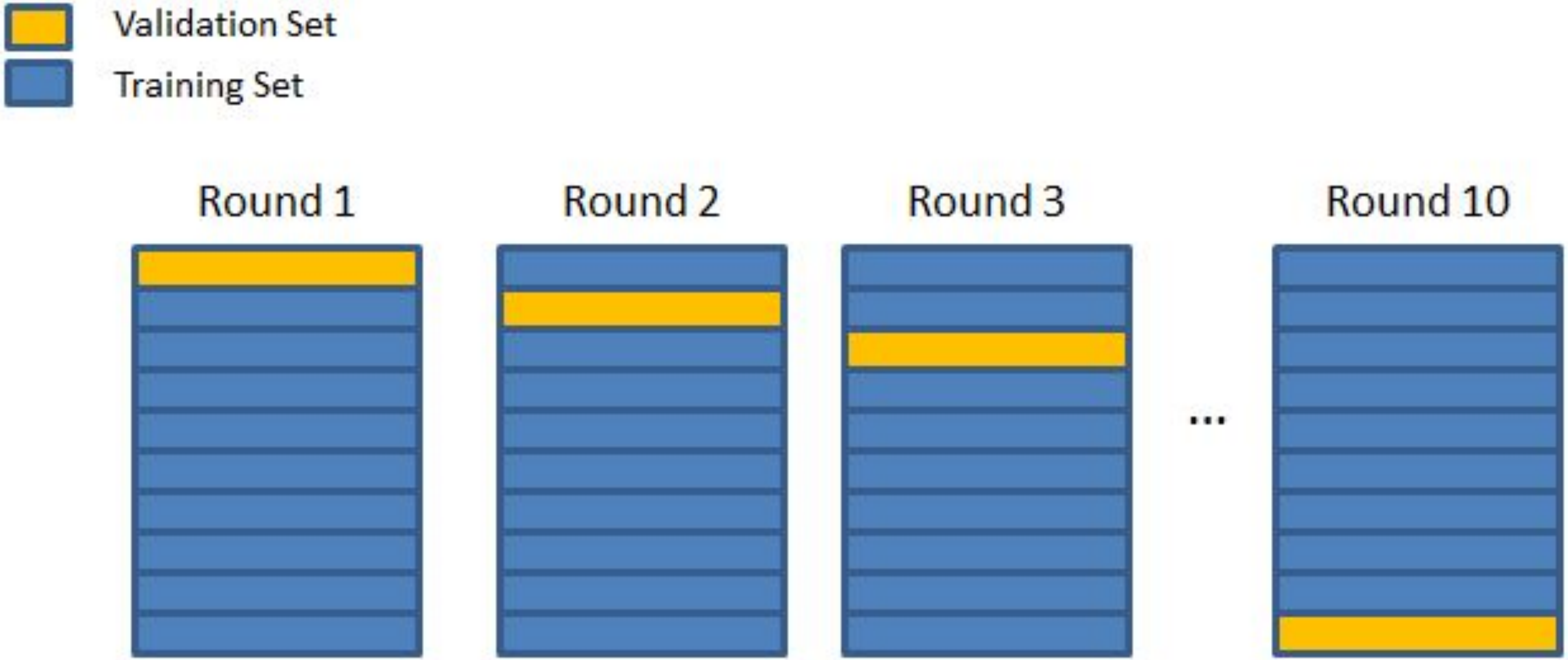


Yes

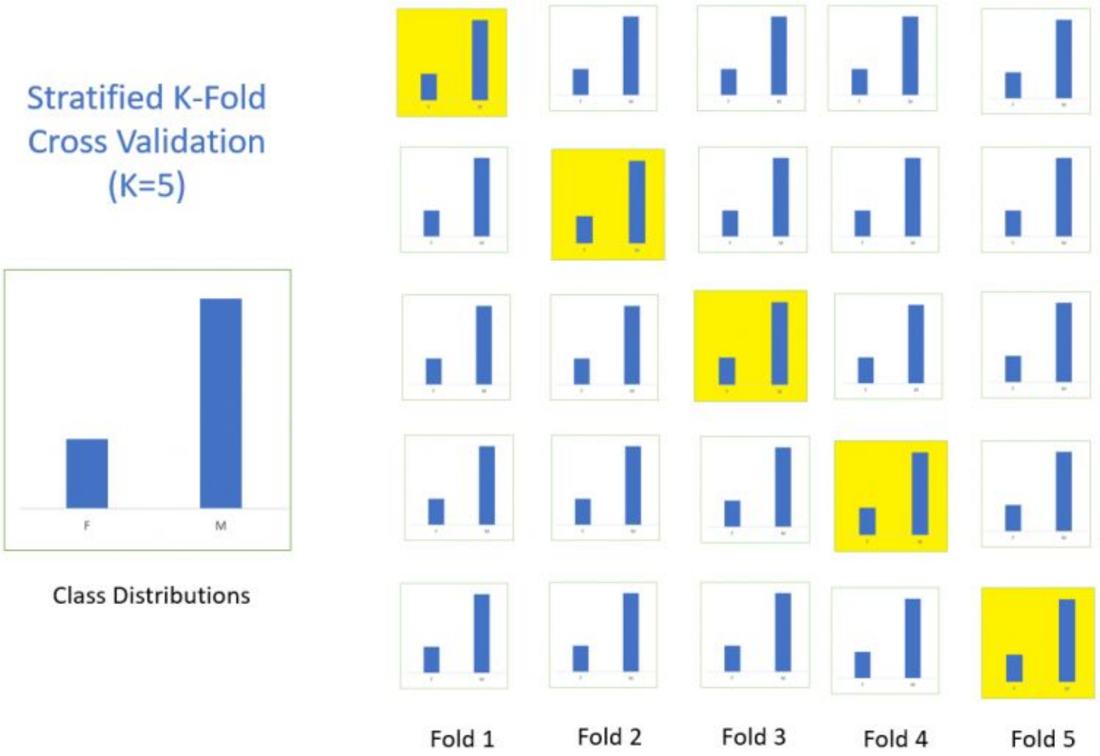


No

Cross Fold Validation



Stratified Cross-Validation



Leave one out / Jack Knife